Live-Music: From Streams to Stages

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From Streams to Stages

- We all listen to music and go to concerts
- We noticed: some artists are huge online but never tour nearby
- This made us curious — does streaming popularity actually predict where artists perform?

→ **Central question:** To what extent do digital streaming metrics predict the physical reality of global concert tours?

Data Sources

Main Sources:

- 🎧 Last.fm API — streaming popularity: listener counts, play counts, geographic reach
- 🎟️ Ticketmaster API — concert data: cities, countries, dates (2023–2026)

Supporting Sources:

- 🌍 RestCountries — country dataset used to classify capital cities
- 🗺️ Spotify Chart Data — weekly chart data used to classify chart vs. non-chart artists

Links:

- https://www.last.fm/api
- https://developer.ticketmaster.com
- https://restcountries.com
- https://charts.spotify.com

Streaming & Ticket Power

→ How strongly is digital popularity on Last.fm related to the scale and intensity of touring?

We investigated:

- Relationship between listener counts and tour scale
- Influence of streaming concentration on touring activity
- Differences in audience size between charting and non-charting artists

https://datascience-project-rchwdmnkmtes77pdt4rhhx.streamlit.app/Streaming_Ticket

Geographic Analysis

→ How do artists choose where to tour - do they revisit cities, play in capitals, and follow their listeners?

We investigated:

- Expansion vs. revisit cities
- Role of capital cities
- Alignment of tours with listeners

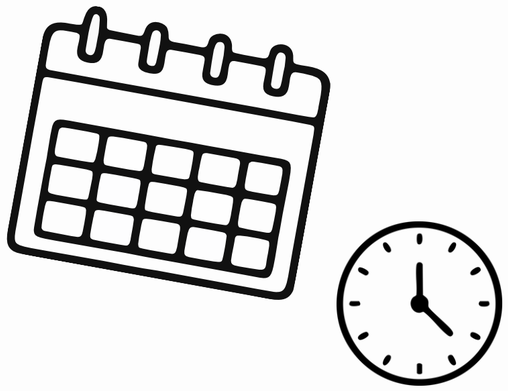
https://datascience-project-rchwdmnkmtes77pdt4rhhx.streamlit.app/Geographic

Market Time & Scheduling

→ How do artists structure their tour schedules and how does this relate to digital popularity?

We investigated:

- Average days between concerts
- Share of concerts scheduled on weekends
- Lead time before concerts



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Conclusion

Question: Does streaming popularity predict touring?
→ Short answer: **yes** — but only partly

What we found out:

- More popular artists play bigger tours — but the difference is smaller than expected
- Artists tour in markets where they have fans — but not always
- A few major hubs dominate: London, Berlin, Dublin
- Many listener markets never get a concert at all

The bigger picture:

Streaming data is a useful signal, but not the whole story
→ Venue availability, management, and budgets matter just as much

What went wrong

→ and how we fixed it

Challenge 1 — Spotify API

- Planned to use Spotify follower counts and popularity scores

Spotify removed these fields in February 2026
→ Switched to Last.fm — turned out to be a better fit anyway
Kept Spotify chart data as a supplementary source

Challenge 2 — Team & Technical

- Merge conflicts and unclear task ownership on GitHub

→ Agreed on a clearer structure and task division

Challenge 3 — Research Questions

- Many of our first questions could not be answered with the available data

→ Multiple revision rounds: Can we measure this? Does it connect to our question?
→ Ended up with 9 questions across 3 themes

💡 **What we would do differently**

- Explore the data first — then define the research questions

Thank you!

Explore our interactive website!

All 9 research questions · Interactive charts · Filterable by genre and popularity

Website Link: https://datascience-project-rchwdmnkmtes77pdt4rhhx.streamlit.app

QR Code

Feel free to ask questions!